

Frame check sequence (FCS)

- * Added at Layer 2.
- * Used for error detection.
- * Helps detect whether the frame became corrupted during transmission.

Network Interface Card (NIC):

NIC perform these tasks

- * Converts bits into frames.
- * Reads the destination MAC address.
- * Checks whether the frame is meant for this device.
- * Removes the Layer 2 header.
- * Passes the packet to Layer 3.

Ethernet Type Field (Layer 2)

Ethernet contains a type field.

Purpose:-

- * It tells the receiving computer, which layer 3 protocol should receive this packet?

e.g. IPv4, IPv6

The NIC reads the type field & forwards the packet to the correct network-layer protocol.

protocol Number (Layer 3)

Inside the IP header is the protocol number.

Purpose:

* It tells Layer 3, which Layer 4 protocol should receive the packet?

e.g. TCP, UDP

port Number (Layer 4)

Inside the TCP/UDP header is the port number.

Purpose:

* It tells the OS, which application should receive the data?

port

80

443

22

23

Application

HTTP

HTTPS

SSH

Telnet

Real - Life Data Transfer Summary :-

- 1) Application creates data
- 2) Transport adds TCP / UDP header.
- 3) Network add IP header.
- 4) Data Link add Ethernet header + FCS.
- 5) Physical converts everything into bits.
- 6) NIC receives bits.
- 7) NIC checks destination MAC
- 8) Layer 2 removes Ethernet header.
- 9) Layer 3 removes IP header.
- 10) Layer 4 removes TCP / UDP header.
- 11) Application receives the original data.